

FE6: Files - structure and access

1. Currently, in general , files are considered as a mere array of bytes.
 - a. Present 2 advantages of this view.
 - b. Try to present at least a disadvantage of this view, giving an example.
2. File naming is essential for its (human) usage, so it generally reflects an organization point of view.
 - a. Explain why, giving an example.
 - b. Reconsider your answer, remembering the way a database is accessed and the way a web search engine is used nowadays.
3. File name extensions are many times very important for accessing files. For each of the following systems, present a couple of examples that show their real importance:
 - Linux
 - MsWindows

(By the way, try to explain why MsWindows, by default, hide file extensions from users.)

4. Consider the following "bin-text" program; compile it and run it (redirecting the output to a file). Take some conclusions on the difference of using binary or text representation of numbers.
(Note: open the redirected file with an hex editor and make sure to read the "hidden" binary number.)

```
#include...
int main()
{
    int i = 1234567;
    write(STDOUT_FILENO, &i, sizeof(i));
    printf("\n%d", i);
    return 0;
}
```

5. Write a program to demonstrate both the **sequential** and the **random** access to a file. Make sure you print the evolution of the *file marker's* value in both situations.
6. Unix makes somewhat of a distinction between a file's **name**, **attributes**, **modes** and **permissions**. Clarify that distinction giving examples.
7. Describe three of the file operations listed on the theoretical classes' supporting sheets that have a similar counterpart in the list of directory operations.
 - a. Why does the operation on directories `readdir()` not have an invocation syntax in Unix similar to the corresponding `read()` operation on files?

8. Write a program that runs through the directory given as a command line parameter, examining the various attributes of all files "contained" in the directory and, before finishing, present the following statistical results:

- i. size of the smallest and largest file found and average and median size of all files;
- ii. number of files in the classes: "read-only", "read and write", "executable";
- iii. minimum and maximum number of files in a directory and average number of files per directory;
- iv. total number of files;
- v. size of blocks used on the disk where the directory resides.

9. Compare, indicating some advantages and disadvantages, the 4 types of implementation of files presented in class, with regard to file block's placement on a disk and steps to access them.

10. The sharing of files by means of symbolic links introduces the need for another "special" file type. Clarify the difference between a (normal) *hard link* to a file and a *symbolic link*. Try to show an advantage of one technique over the other.

11. Compare the two methods of managing the free blocks of a disk that were presented in class: by "linked list of blocks" and by "bit table".

- a. Present some clarifying calculations for a 500 GiB disk with 4 KiB blocks.